Bodan Arsovski

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JOBS

1.25 – 7.26 Senior research fellow at *University College London.*

8.24 – 9.24 Research fellow at *University College London.*

9.23 – 7.24 Member of the *Institute for Advanced Study* in Princeton, New Jersey.

8.22 – 8.23 Research fellow at *University College London.*

8.21 – 11.21 Research associate at the *University of Sheffield.*

SCHOOLS

11.14 – 3.19 PhD in mathematics from *Imperial College London.*

I was awarded the "President's PhD scholarship", which is a competitive scholarship that fully funded my PhD studies.

10.10 – 6.14 BSc & MMath in mathematics from the *University of Oxford.*

I graduated with "First-class honours" in all four years, which is the top possible grade in the UK system. I received the "Junior mathematical prize" for ranking top 3 in my generation on the final exams in mathematics. I was awarded the "Reach Oxford scholarship", which is a competitive scholarship that fully funded my undergraduate studies. I also received the "Clothworkers' scholarship", which is a prize awarded by St. Catherine's College.

ARTICLES

- "The p-adic Kakeya conjecture"
J. Amer. Math. Soc. **37** (2024), 69–80

The classical *Kakeya conjecture* asks if all compact subsets of $\mathbb{R}^n$ containing a line segment of unit length in every direction have full Hausdorff dimension. In this article I prove the natural analogue of the classical Kakeya conjecture over the p-adic numbers — more specifically, that all compact subsets of $\mathbb{Q}_p^n$ containing a line segment of unit length in every direction have full Hausdorff dimension — a conjecture which was first discussed in the 1990s by James Wright. In fact, more generally I prove the p-adic analogue of the *Kakeya maximal conjecture,* which is a functional version of the Kakeya conjecture proposed by Jean Bourgain in the 1990s.

The reason this is interesting is that the Kakeya conjecture boils down to the highly combinatorial question of packing thin tubes as tightly as possible, and in all known cases tubes can be packed exactly as tightly in $\mathbb{Q}_p^n$ as in $\mathbb{R}^n$ (because the ways in which thin tubes can intersect are similar in both settings).

This result was featured in *Quanta Magazine* in the articles "A question about a rotating line helps reveal what makes real numbers special" and "The year in math".

- "On the Minkowski dimension of certain Kakeya sets"
  Mat. Bilten **46** (2022), 77–82
- "Limiting measures of supersingularities"
  Preprint
- "Local irreducibility on the eigencurve"
  Preprint
- "On the reductions of certain two-dimensional crystalline representations, II"
  Preprint
- "On the reductions of certain two-dimensional crystalline representations"
  Doc. Math. **26** (2022), 1929–1979
- "On the reductions of certain two-dimensional crystabelline representations"
  Res. Math. Sci. **8** (2022), #12
- "Additive bases via Fourier analysis"
  Combin. Probab. Comput. **30** (2021), 930–941
- "A proof of Snevily's conjecture"
  Israel J. Math. **182** (2011), 505–508

TEACHING

- *UCL MATH0014:* I have taught problem classes in linear algebra for second year undergraduates at *University College London.* These were classes in a typical classroom setting, for an audience of ≈20.
- *UCL MATH0008:* I have covered lectures in applied mathematics for first year undergraduates at *University College London.* These were lectures in a lecture hall setting, for an audience of ≈200.